

NAMMATUTOR TOPPER NOTES

Life Processes

— and the story of Nutrition —

CLASS 10 · BIOLOGY · CHAPTER 1

What is a Life Process?

Autotrophic Nutrition

Photosynthesis

Heterotrophic Nutrition

Nutrition in Amoeba

Human Digestive System

1 What is a Life Process?

the maintenance functions every living thing must keep running

DEFINITION

Life processes are the basic functions performed by organisms to **maintain their life** — even when the body is not doing anything else visible. A resting person is still nourishing cells, exchanging gases, pumping blood and removing waste.

Nutrition

Respiration

Transportation

Excretion

Why "maintenance" is the key word

Growth and movement can pause — a seed can lie dormant for years. But nutrition, respiration, transportation and excretion cannot stop even for a minute without damaging the organism. That is what separates a life process from an ordinary activity.

2 Nutrition — the first process

how an organism gets and uses food

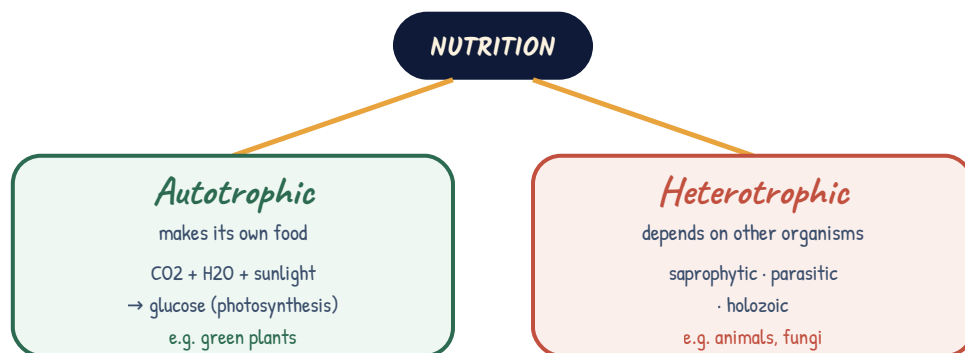


Fig 1.1 — the two routes to getting food

ONE-LINE RECALL

"Green plants **COOK** their food. Animals **COLLECT** it."

Why living things can't skip any of these

Process	What happens if it stops
Nutrition	No raw material for energy or growth — the organism starves
Respiration	No energy released from food — cells cannot function
Transportation	Nutrients and oxygen never reach the cells that need them
Excretion	Toxic waste builds up inside the body — cells get poisoned

3 Autotrophic Nutrition

photosynthesis – the food-making reaction inside a leaf

THE EQUATION

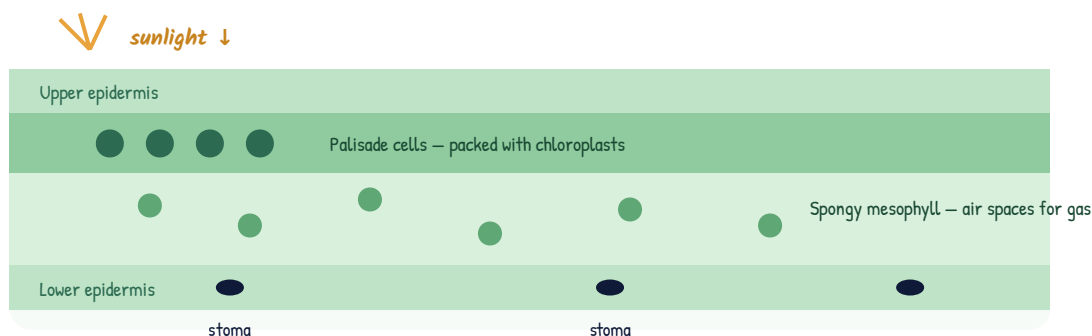
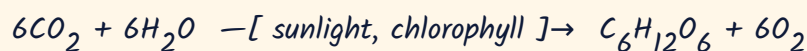
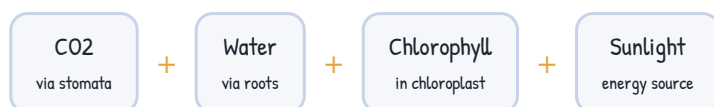


Fig 1.2 – cross-section of a leaf, the site of photosynthesis

Four raw materials



BOARD TRAP

Photosynthesis has TWO stages: the **light reaction** (splits water, releases O₂, needs light directly) and the **dark reaction** (fixes CO₂ into glucose, does NOT need light directly — it needs the products of the light reaction). Examiners love asking which stage a given event belongs to.

Light reaction vs Dark reaction – side by side

Light reaction	Dark reaction
Occurs in the grana (thylakoid membranes)	Occurs in the stroma
Water is split — oxygen released as a by-product	Carbon dioxide is fixed into glucose
Directly needs sunlight	Uses ATP and NADPH made by the light reaction — no light required directly
Also called the photochemical phase	Also called the biosynthetic / Calvin cycle phase
Grana → light reaction	Stroma → dark reaction
Both happen inside the same chloroplast	

4 Heterotrophic Nutrition

three ways of depending on someone else's food

Type	How it feeds	Example
Saprophytic	Secretes digestive juices onto dead matter, then absorbs the dissolved nutrients	Bread mould, mushroom
Parasitic	Derives nutrition directly from a living host, usually harming it	Cuscuta, tapeworm
Holozoic	Takes in solid/liquid food, then digests it internally	Amoeba, human beings

Holozoic nutrition – five steps, every time

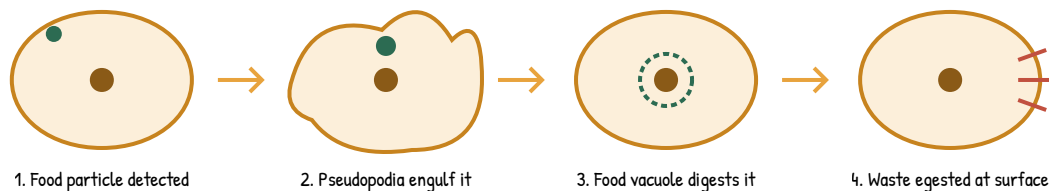
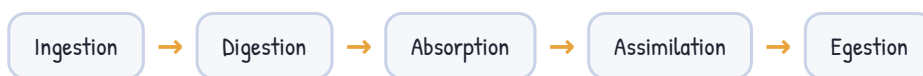


Fig 1.3 — Amoeba feeding by phagocytosis (a moving, engulfing cell)

REMEMBER

Amoeba has no fixed mouth. Any part of its cell surface can form **pseudopodia** ("false feet") to surround and trap food — this is exactly what your white blood cells do to bacteria in your own body.

Saprophyte or parasite – the one-question test

Ask: is the source of food alive? A saprophyte feeds on dead, decaying matter — the organism it feeds on is already dead. A parasite feeds on a living host, and usually stays attached to it for its whole feeding life. Both are heterotrophic, but only one harms a living organism directly.

ONE-LINE RECALL

"Saprophytes clean up the dead. Parasites drain the living."

5 Nutrition in Human Beings

one long tube, four digestive glands, one job

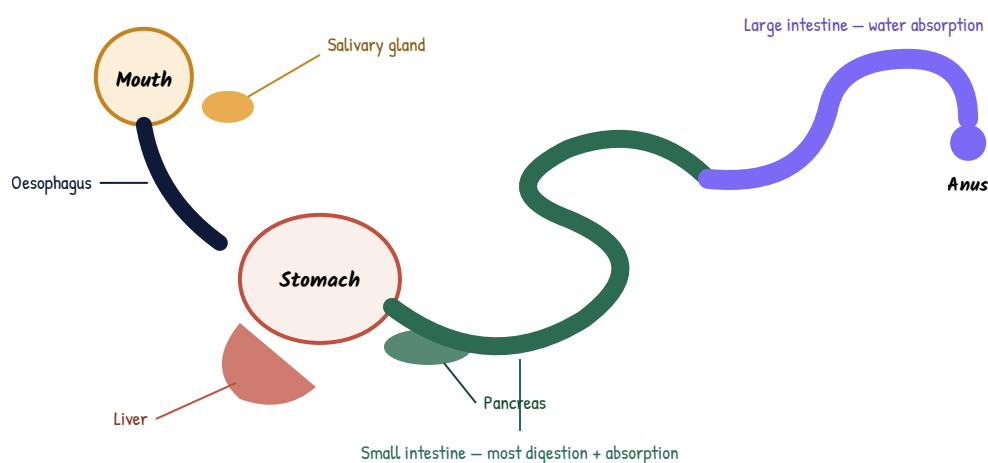


Fig 1.4 — the human alimentary canal (simplified)

What each gland adds

Gland	Secretion → does what
Salivary	Saliva — starch → sugar
Liver	Bile — emulsifies fat
Pancreas	Enzymes for all 3 food groups

Stomach's own job

Gastric glands release HCl (kills germs, activates pepsin) and pepsin (begins protein digestion). Mucus lines the wall so the stomach doesn't digest itself.

EXAM-LINE WORTH MEMORISING

"Most digestion AND most absorption both finish in the small intestine — that's why it's the longest part of the canal."

Digestive enzymes at a glance

Enzyme	Made in	Acts on	Produces
Salivary amylase	Salivary glands	Starch	Maltose
Pepsin	Stomach (gastric glands)	Proteins	Peptones
Trypsin	Pancreas	Proteins / peptones	Peptides
Lipase	Pancreas	Fats (emulsified by bile)	Fatty acids + glycerol